

Sliding Window Distinct Values

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Your task is to calculate the number of distinct values in each window of k elements, from left to right.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n - k + 1$ values: the numbers of distinct values.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
8 3 1 2 3 2 5 2 2 2	3 2 3 2 2 1